CLOSED POINTS WITH FINITE SEPARABLE RESIDUE FIELD IN A NONEMPTY OPEN OF AFFINE SPACE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is complete as written: the reduction to a principal open subset, the contradiction hypothesis, and the infinitude of the separable closure are all present in the source.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-affine-space-over-field is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Separable closure k^sep and its infinitude. Standard field theory, used at proof lines 4652-4654. Supplies the infinite field of definition points on which f would have to vanish.
- A polynomial in d variables over an infinite field does not vanish on all d-tuples. Standard algebra, used implicitly at proof line 4654. Supplies the contradiction that finishes the proof.
- Closed points of A^d_k and their residue fields. Standard, a point of (k^sep)^d is a closed point whose residue field is a finite separable extension of k. Supplies that a point of (k^sep)^d lying outside V(f) is exactly the w the statement asks for.

Lemma 0.1. *Let k be a field. Let $d \geq 0$. Let $W \subset \mathbf{A}^d_k$ be nonempty open. Then there exists a closed point $w \in W$ such that $k \subset \kappa(w)$ is finite separable.*

Proof. After possible shrinking W we may assume that $W = \mathbf{A}^d_k \setminus V(f)$ for some $f \in k[x_1, \ldots, x_d]$. If the lemma is wrong then $f(a_1, \ldots, a_d) = 0$ for all $(a_1, \ldots, a_d) \in (k^{sep})^d$. This is absurd as k^{sep} is an infinite field. □

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